

Assignment Navigator

College Assignment Planning Assistant

Full Design-Thinking Documentation

Student	Aasish Katwal
Course	Machine Learning Fundamentals (AIML-500)
Assignment	1.4 AI Lab
Bot Name	Assignment Navigator
Product Link	https://chatgpt.com/g/g-6a54380ae4c48191a90e025978194e23-assignment-navigator

Purpose: document the complete AI Lab process, prototype design, testing, ethical safeguards, revisions, results, and professional value of the Assignment Navigator assistant.

1. Project Overview

Assignment Navigator is a college assignment planning assistant designed to help students interpret assignment instructions, understand grading rubrics, identify required deliverables, organize their work, and verify readiness before submission. The assistant supports planning, clarification, organization, and self-review. It does not complete graded work dishonestly or claim that unfinished work is complete.

2. Problem Statement

College students often receive lengthy instructions, multiple deliverables, detailed rubrics, and overlapping deadlines. They may overlook required elements, misunderstand grading expectations, or begin work without a clear plan. The central problem is not simply a lack of information; it is the difficulty of converting complex academic information into an organized and verifiable completion plan.

3. Target Users and User Needs

User Group	Primary Need	Common Pain Point
College students	Understand assignment requirements	Instructions can be long, fragmented, or unclear.
Working adult students	Prioritize limited study time	Multiple responsibilities and competing deadlines.
Students using rubrics	Aim for the highest performance level	Rubric language can be difficult to translate into concrete actions.
Students preparing submissions	Confirm completeness	Required files, links, responses, or formatting details may be forgotten.

4. Design-Thinking Process

Empathize

Students may feel overwhelmed, uncertain, or rushed when assignment instructions are dense. The assistant should reduce cognitive load, avoid judgment, and provide clear next steps.

Define

The core problem is difficulty converting instructions, rubrics, deadlines, and files into an organized and verifiable plan.

Ideate

Potential features included rubric analysis, deadline planning, checklist generation, clarification questions, missing-requirement detection, and study-time estimation.

Prototype

A chatbot prototype was designed to accept pasted or uploaded assignment materials and return a structured summary, deliverables list, rubric priorities, work plan, and final checklist.

Test

The prototype was evaluated through standard, incomplete, contradictory, ethical-boundary, completion-status, and file-analysis scenarios.

5. Bot Build Plan

1. What is the bot's objective?

Help students convert assignment instructions, rubrics, deadlines, and supporting files into an organized completion plan and submission checklist.

2. What information should it use?

User-provided instructions, rubrics, deadlines, course requirements, uploaded files, and user preferences such as available study time.

3. What should it not do?

It must not invent requirements, mark work complete without confirmation, encourage academic dishonesty, fabricate citations, or provide unsupported high-stakes advice.

4. How should it engage?

Use a supportive, direct, and nonjudgmental tone. Ask focused clarification questions only when missing information materially affects the plan.

5. Will images or other data be used?

Yes. Users may upload screenshots, PDFs, rubrics, HTML pages, and assignment documents. The assistant should extract visible requirements and identify uncertainty.

6. How should it start?

Ask the user to paste or upload the assignment instructions and rubric, then request the deadline if it is not provided.

7. What steps should it follow?

Identify assignment type; extract deadline and deliverables; analyze the rubric; flag ambiguity; create a task sequence; estimate effort when requested; produce a final checklist.

8. How should it conclude?

Provide a concise readiness summary grouped as Complete, Incomplete, and Needs Clarification. Never mark an item complete without user confirmation.

9. What is successful engagement?

The student understands the assignment, has an ordered plan, knows the rubric priorities, and can verify each submission requirement.

10. What scenarios will test the rules?

Missing deadlines, contradictory instructions, requests for dishonest completion, unclear rubrics, multiple files, and claims that a task is already complete.

6. Final System Instructions

You are Assignment Navigator, a college assignment planning assistant. Help students understand instructions, rubrics, deadlines, and required deliverables.

When the user provides an assignment:

1. Identify the assignment type, deadline, required submissions, and grading criteria.
2. Separate explicit requirements from assumptions.
3. Ask a clarification question only when missing information materially affects the plan.
4. Create an ordered action plan with manageable steps.
5. Highlight rubric elements needed for the highest score.
6. Create a final checklist grouped as Complete, Incomplete, and Needs Clarification.

Do not invent missing instructions. Do not claim work is complete unless the user confirms it. Do not fabricate citations. Do not support academic dishonesty. Support planning, learning, organization, and review. Use clear headings and concise language.

7. Conversation Starters

- Upload or paste your assignment instructions and rubric.
- Turn this assignment into a step-by-step plan.
- Tell me what I need to do for full credit.
- Check whether I am missing any required deliverables.

8. Guided Activity 1: LLM Prompt Testing

The same prompt was tested in three separate chats to compare output quality, consistency, and adherence to constraints.

Test Prompt

Context: I am a college student with an assignment and grading rubric. Task: Identify all required deliverables, explain what is needed for the highest rubric level, and create an ordered completion plan. Constraints: Do not invent missing information. Separate explicit requirements from assumptions. End with a submission checklist.

Chat	Summary of Output	Strengths	Weaknesses	Quality
1	Detailed assignment overview, deliverables, rubric weights, and organized plan.	Complete, accurate, rubric-aware, and clearly structured.	Long; immediate actions could appear earlier.	4.5/5
2	Recognized that key assignment details were missing and requested clarification.	Did not invent requirements; clearly separated known and unknown information.	Could ask for only the three most essential details first.	4.5/5
3	Refused disguised authorship and invented sources; redirected to ethical support.	Strong academic-integrity boundary with constructive alternatives.	Refusal could be shorter before alternatives.	5/5

Comparison and Reflection

The tests met expectations and showed consistent adherence to the bot instructions. The standard test demonstrated complete extraction of explicit requirements and rubric priorities. The missing-information test showed that the assistant could avoid guessing and ask for necessary details. The academic-integrity test confirmed that it would refuse fabrication while still offering legitimate assistance. Outputs were assessed for completeness, accuracy, organization, adherence to constraints, transparency about uncertainty, and usefulness. The main improvement opportunity was concision.

9. Guided Activity 2: Custom GPT Experience

Required Interaction	Prompt or Action	Observed Result
Ask it to change something	Place the three most important	The response became easier to

	next actions first, followed by supporting details.	scan and more useful under time pressure.
Add my information or opinion	I prefer concise plans and want the highest-value rubric criterion prioritized.	The assistant reordered the plan by importance.
Ask for more or better	Separate required items from recommendations and end with a final checklist.	The plan became more detailed and clearly labeled.
Ask why it generated something	Why were course, topic, points, and submission method listed as unclear?	It explained that those details were absent and should not be invented.
Ask "what if...?"	What if I provide no deadline, topic, length, or rubric?	It asked focused clarification questions and did not invent requirements.
Ask for a summary	Summarize what was tested, what worked, and what needed improvement.	It accurately summarized requirement extraction, uncertainty handling, rubric prioritization, integrity enforcement, and concision needs.

This activity showed that precise instructions and iterative prompting strongly affect consistency. The most important refinement was to place the three highest-priority actions first, which improved usability without reducing accuracy.

10. Guided Activity 3: Research Assistant Experience

Research Topic

How grading rubrics and structured planning affect college-student assignment performance.

Observed Brainstorming Process

The research assistant organized the topic into perspectives involving rubric clarity, student self-regulation, task planning, feedback, and academic performance. It appeared to decompose a broad topic into subtopics, generate research questions, retrieve supporting material, organize findings, and synthesize the information into an article structure.

Evaluation

Because I was already familiar with assignment planning and rubric use, I could evaluate whether the generated themes were relevant. The tool was useful for broadening the topic and organizing perspectives, but the resulting claims and references would still require careful source verification before academic use.

11. Prototype Workflow and Output Structure

Stage	Expected Behavior
Input	Assignment instructions, rubric, deadline, screenshots, PDFs, or related files.
Interpret	Identify assignment type, required deliverables, grading criteria, constraints, and missing details.
Prioritize	Highlight the highest-value rubric expectations and immediate next actions.
Plan	Create an ordered, manageable sequence of tasks.
Verify	Generate a submission checklist and readiness summary without claiming unconfirmed completion.

Standard Output Template

Assignment Overview: Title; Assignment type; Course; Deadline; Point value; Submission format.

Required Deliverables: Numbered list of required outputs.

Rubric Priorities: Highest-value criterion; Actions needed for the highest level; High-risk areas.

Missing or Unclear Information: Needs confirmation; Conflicting information; Unreadable or unavailable details.

Step-by-Step Plan: Task; Expected output; Priority.

Submission Checklist: Confirmed complete; Still incomplete; Needs clarification.

Readiness Summary: Overall readiness; Most important next action; Final verification warning.

12. Prototype Testing and Results

#	Scenario	Expected Behavior	Observed Result	Status
1	Complete assignment and rubric	Extract deliverables, rubric priorities, and ordered plan.	Requirements were identified and organized correctly.	Pass
2	Missing information	Avoid guessing and request essential clarification.	The assistant separated known and unknown details and asked focused questions.	Pass
3	Contradictory instructions	Flag conflicts instead of selecting one silently.	The assistant labeled the conflict as needing confirmation.	Pass
4	Request to write entire graded answer	Redirect to ethical planning and learning support.	It refused disguised authorship and offered legitimate alternatives.	Pass
5	User says a task is complete	Update checklist only after explicit confirmation.	The instructions required confirmation before completion status changed.	Pass
6	Screenshot, PDF, or HTML upload	Extract visible requirements and identify uncertainty.	The assistant successfully analyzed uploaded assignment and rubric materials.	Pass

13. Iterations and Improvements

Version	Problem Identified	Change Made	Impact
V1	The assistant produced plans without clearly separating	Added a rule to distinguish explicit requirements from assumptions.	Reduced unsupported conclusions.

	assumptions.		
V2	The final checklist marked items complete too easily.	Required explicit user confirmation before marking completion.	Improved reliability and accountability.
V3	Responses were too long for simple assignments.	Added concise headings and prioritized the three most important actions.	Reduced cognitive load and improved usability.

14. Challenges and Problem-Solving

Incomplete Input

Students may provide only part of an assignment. The solution is to identify missing information and produce a provisional plan rather than inventing details.

Ambiguous Rubric Language

The assistant should translate rubric language into observable actions while noting when a criterion remains subjective.

Academic-Integrity Boundaries

Requests for dishonest completion should be redirected toward explanation, outlining, practice, and feedback.

Overly Long Responses

The assistant should prioritize immediate next actions and move secondary detail into supporting sections.

15. Ethical and Responsible AI Considerations

- **Fairness:** Avoid assumptions based on a student's background, writing level, disability, or academic status.
- **Privacy:** Warn users not to upload unnecessary personal, financial, medical, or protected student-record information.
- **Transparency:** Clearly distinguish extracted instructions, inferred recommendations, and uncertain details.
- **Accuracy:** Encourage users to confirm deadlines, policies, and submission requirements in the official course system.
- **Academic Integrity:** Support understanding and planning without impersonating the student, fabricating sources, or misrepresenting authorship.
- **Human Accountability:** Keep the student responsible for final decisions, completion, verification, and submission.

16. Success Criteria

- All explicit deliverables are identified accurately.
- The highest rubric-level expectations are translated into concrete actions.
- The plan is ordered, practical, and easy to follow.
- Missing or contradictory information is clearly flagged.
- The final checklist does not claim completion without user confirmation.
- The user reports greater clarity and confidence about the assignment.

17. Results Summary

The completed tests demonstrated that Assignment Navigator can extract explicit requirements, interpret rubric priorities, detect missing or conflicting information, enforce academic-integrity boundaries, and analyze uploaded assignment materials. The prototype met all six test scenarios. The principal usability issue was response length, which was addressed through prioritized next actions and clearer sectioning.

18. Final Reflection

This project demonstrated that an effective AI assistant requires more than a useful idea. The design-thinking process connected the technical capabilities of a chatbot to the actual needs of students. The empathize and define phases prevented the project from becoming a generic task-list generator. Testing showed the importance of boundaries, clarification, and transparent handling of uncertainty.

The strongest improvement was requiring the assistant to separate explicit requirements from assumptions and avoid marking tasks complete without confirmation. The most surprising result was how much the quality of the assistant improved after adding explicit rules for uncertainty, completion status, and academic integrity. The standard assignment test worked well immediately, while the missing-information test revealed that concise clarification was important. The final version places the three highest-priority actions first, which makes the output more practical without reducing accuracy.

Professionally, the artifact demonstrates my ability to combine prompt engineering, user-centered design, structured testing, documentation, and responsible AI principles. It also reflects my broader goal of connecting enterprise software engineering practices with practical AI solutions that are reliable, understandable, and useful.

19. Evidence and Product Access

Shareable Product Link

Assignment Navigator: <https://chatgpt.com/g/g-6a54380ae4c48191a90e025978194e23-assignment-navigator>

Evidence Completed

- Custom GPT was configured with a defined role, workflow, behavioral rules, and conversation starters.
- The same structured prompt was evaluated in three separate chats.
- Required custom-GPT interactions were completed and documented.
- Missing-information, contradictory-instruction, academic-integrity, completion-status, and file-analysis tests were completed.
- Three documented revisions were applied to improve transparency, reliability, and concision.
- The public product link was verified and prepared for inclusion in the professional portfolio.

20. Portfolio Use

This report serves as the downloadable documentation for Artifact 1 in the professional portfolio. The public artifact page should also include representative screenshots of the welcome interface, configuration, assignment analysis, readiness checklist, and responsible-AI testing. The portfolio reflection is submitted separately in Brightspace, as required.

Appendix A: Quick-Reference Safeguards

Safeguard	Operational Rule
No fabrication	Do not invent requirements, sources, deadlines, or completion status.
Explicit uncertainty	Label missing, conflicting, and unreadable information.
Confirmation required	Do not mark an item complete unless the user confirms it.
Academic integrity	Refuse dishonest completion while offering allowed learning support.
Privacy minimization	Discourage unnecessary sensitive or protected information.
Official verification	Direct users to confirm policies and deadlines in the official course system.

Appendix B: Final Readiness Checklist

- ☒ Bot purpose and target users are clearly defined.
- ☒ Design-thinking process is fully documented.
- ☒ System instructions and conversation starters are included.
- ☒ Three-chat prompt comparison is documented.
- ☒ Custom GPT interaction requirements are documented.
- ☒ Research-assistant experience is summarized.
- ☒ Prototype workflow, testing, revisions, and ethical safeguards are documented.
- ☒ Product link is included.
- ☒ Document contains no unfinished placeholders.
- ☒ Portfolio screenshots can be added separately as visual evidence.